

Intelligent Financial Fraud Detection: technical report

DLW Datathon 2026, Track 2 · Team TEAM_NAME · Notebook: fraud_detection_track2.ipynb · Model: model.pkl

1. Our understanding of the problem

20,000 labelled transactions, 353 fraudulent (1.8%); 12,000 to score. Accuracy is meaningless (98.2% by predicting "never fraud"), so the job is twofold: **rank** fraud above legitimate transactions as well as possible (PR-AUC), then **choose a threshold** that trades missed fraud against disturbed customers, which is the business question in the problem statement. Only 353 positives means every estimate is noisy and flexible models will memorise noise; the hidden test set has twice the share of missing values, so gaps must be handled, not dropped.

2. Data preparation and exploration

Quality. No duplicate ids, no impossible values. Eight columns have 0.5–0.7% missing in training and 2.0–2.4% in the test set. We keep every row: trees receive NaN natively, the linear model gets a median fill, and each column gets a *missing* indicator because absence is itself a weak signal (fraud rate 3.0% where `spend_last_24h` is missing).

What separates fraud. A new device (7.5% vs 1.2%); merchant category (luxury 5.3%, cash transfer 4.7%, electronics 3.9%, grocery 0.8%); any non-SG country (3–4% vs 1.3%); bank transfer and e-commerce channels; 23:00–03:00; larger amounts (median S\$224 vs S\$74); younger accounts (336 vs 796 days); bursts of activity in the last hour and day. Each effect is modest; none identifies fraud alone.

Features (31, all from one row). Raw columns;

* `_missing` flags; log amount and log account age; the payment relative to the account's recent average ticket; its share of the day's spend; the share of the day's transactions in the last hour; explicit flags for night, foreign, new account and high-risk category; and two interactions, new device × foreign and new device × high-risk category. The same function runs at training and scoring time.

Validation. Stratified 5-fold cross-validation with out-of-fold probabilities for every row. PR-AUC is the primary metric; F1 and recall are read at a threshold chosen on those out-of-fold scores, never at 0.5.

3. Models compared (cross-validated)

Model	PR-AUC	ROC-AUC	Best F1	Note
Logistic regression, raw columns (baseline)	0.191	0.772	0.272	strong for its simplicity
Logistic regression, engineered features, $C = 0.03$	0.201	0.781	0.288	features add signal
HistGradientBoosting (sklearn, 15 leaves)	0.166	0.752	0.258	over-fits
LightGBM, large (15 leaves, 600 trees)	0.179	0.733	0.266	worse than the baseline
LightGBM, small (3 leaves, 200 trees, strong regularisation)	0.205	0.785	0.291	learns a few useful splits
Final: soft-vote blend of the two best	0.206	0.785	0.290	different mistakes, so averaging helps

Why this model. The signal is weak and the positives are few, so capacity hurts: the big boosted model lost to a linear baseline. A heavily regularised linear model and a 3-leaf LightGBM each reach the ceiling of this data by different routes; blending them is the most robust choice we found, and both halves are standard scikit-learn/LightGBM objects, so the model loads anywhere with `joblib`.

4. Operating point

We maximise $F_{1.5}$ rather than F_1 , weighting recall slightly above precision because a missed fraud normally costs more than a false alarm a customer can clear with one tap. At the chosen threshold (0.554, out-of-fold): **recall 34%, precision 20%, F1 0.25, 3.1% of transactions flagged**; 121 of 353 frauds caught, 494 legitimate customers asked to confirm out of 19,647 (2.5%). Maximising F_1 instead gives F1 0.29 with recall 25%. The threshold is baked into the saved model (`FixedThresholdClassifier`), so `predict()` applies it and `predict_proba()` gives the ranking score. On the public test set the model flags 4.4%, a little more than in training, consistent with its higher missingness.

5. Assumptions, limitations, next steps

- Assumptions:** the hidden set comes from the same process; rows are independent (no customer id); the 24-hour and 1-hour behavioural counts were computed before the transaction, so they are not leakage.
- Limitations:** PR-AUC ≈ 0.21 means roughly one in five top-ranked flags is real; that is a property of the available columns, and every model family we tried lands in the same band. With 353 positives the threshold itself moves by a few points of recall between folds.
- Next:** customer or card identifiers for true behavioural baselines (where production systems get most of their lift); cost-based thresholding with real loss and friction figures; probability calibration for a tiered response (allow, confirm by SMS, block); monitoring of the missingness rate and country mix, which already differ between train and test.